Write the CFG for the following languages

TOC Unit-3

① $L = \{0^n 1^n \mid n \geq 1\}$

② $L = \{a^i b^j c^k : i+j=k, i \geq 0, j \geq 0\}$

③ $L = \{a^n b^m c^k : n+2m=k\}$

④ $L = \{0^m 1^m 2^n : m \geq 0, n \geq 0\}$

⑤ $L = \{a^i b^j \mid i \neq j, i \geq 0, j \geq 0\}$

⑥ $L = \{a^n b^{n-3} \mid n \geq 3\}$

⑦ $L = \{w w^R \mid w \in \{a,b\}^*\}$

⑧ palindromes over $\{0,1\}^*$

⑨ $L = \{0^n 1^n : n > 0\}$

⑩ $L = \{a^n b^{2n}$ over $\Sigma = \{a,b\} n \geq 1\}$

⑪ all strings over $\{a,b\}$ that are even & odd palindrome

⑫ $L = \{a^n w w^R b^n \mid w \in \{0,1\}^*$ and $n \geq 2\}$

⑬ $L = \{a^k b^m c^n \mid m+n=k$ & $m,n \geq 1\}$

⑭ $L = \{w \in \{a\}^* \mid |w| \bmod 3 \neq |w| \bmod 2\}$

⑮ $L = \{$String of a's & b's with equal no of a's & b's$\}$

⑯ $L = \{a^n b^m \mid m+n$ is even$\}$

⑰ $L = \{a^n b^m \mid m \geq 1, n \geq 1, nm \geq 3\}$

⑱ $L = \{w : |w| \bmod 3 = 0$ where $w \in (a,b)^*\}$

(9) $\{w \mid n_a(w) = n_b(w)\}$

(10) $(011+1)^*(01)^*$

(11) $L = \{w : w \text{ is a palindrome, where } w \in (a,b)^*\}$

2) $L = \{w : n_a(w) \bmod 2 = 0 \text{ where } w \in (a,b)^*\}$

3) $L = \{a^n b^{2n} \mid n \geq 1\}$

4) $L = \{a^i b^j \mid 2i = 3j+1\}$

5) $L = \{0^{n+2} 1^n \mid n \geq 1\}$

6) Grammar to generate integers & write derivation for unsigned integer 1965

7) $L = \{ab(bbaa)^n bba(ba)^n : n >= 0\}$

26. $E \to E+E \mid E-E \mid E*E \mid E/E \mid (E) \mid a$
where (E) is the start symbol. Find unambiguous grammar.

27. Convert the Below grammar to CNF
$S \to aBa \mid abba$
$A \to ab \mid AA$
$B \to aB \mid a$

28. Show that following grammar is ambiguous
$S \to aB \mid bA$
$A \to aS \mid bAA \mid a$
$B \to bS \mid aBB \mid b$

29. Show that following grammar is ambiguous
$S \to a \mid Sa \mid bSS \mid SSb \mid SbS$

30. Convert the following to CNF

(a) $S \to aB \mid bA$
$A \to a \mid aS \mid bAA$
$B \to b \mid aS \mid aBB$

(b) $S \to aACa$
$A \to a \mid B$
$B \to C \mid c$
$C \to cC \mid E$

31. Simplify:
$S \to AB$
$A \to a$
$B \to C \mid b$
$C \to D$
$D \to E \mid bC$
$E \to d \mid Ab$

32. Explain the hierarchy within the class of CFL.

33. Show that the language given is inherently ambiguous.

$L = \{a^n b^n c^m \mid n, m \geq 0\} \cup \{a^n b^m c^n \mid n, m \geq 0\}$

34. Prove that $L = \{a^n b^n c^n \mid n \geq 0\}$ is not CFG.